ONLINE WORKSHOP · PARTICIPANT GUIDE

Kadi-APY for everyday lab work

From lab files to findable data with Kadi4Mat and a little Python. No programming experience needed. Please work through **page 1 before the workshop**; it takes about 20 minutes.

DATE & TIME	MEETING LINK	HELP BEFORE THE DAY
[FILL IN]	[FILL IN]	[FILL IN e-mail]

Before the workshop: checklist

- [ ] **1. Log in to Kadi4Mat** 2 min
  Open **kadi4mat.iam.kit.edu** and log in. Kadi4Mat is the research data repository where our data will be stored.

- [ ] **2. Create a personal access token** 3 min
  A token is like a password for programs. In Kadi4Mat:
  1. Click your name (top right) → **Settings** → **Access tokens**.
  2. Name it `workshop`, set an expiry date a few days after the workshop, and create it.
  3. **Copy the token right away** into a text file. It is shown only once.

- [ ] **3. Check your Google account** 1 min
  We use **Google Colab**: Python in your web browser, free, nothing to install. Open **colab.research.google.com** and sign in with any Google account.

- [ ] **4. Open the workshop notebook and save a copy** 2 min
  Open [FILL IN short link], then **File → Save a copy in Drive**. Work in your copy from now on.

- [ ] **5. Store your token as a Colab secret** 2 min
  In the notebook, click the **key icon** in the left sidebar → **Add new secret**. Name: `KADI_PAT` (exactly like this), Value: your token. Switch on **Notebook access**.

- [ ] **6. Run the setup cells (section 0)** 3 min
  Click into the first code cell and press `Shift` + `Enter`, one cell after the other. In 0.4 change `"xy"` to your initials. You are done when you see three green ticks: **Ready**, **Connected**, **Your prefix is ...**

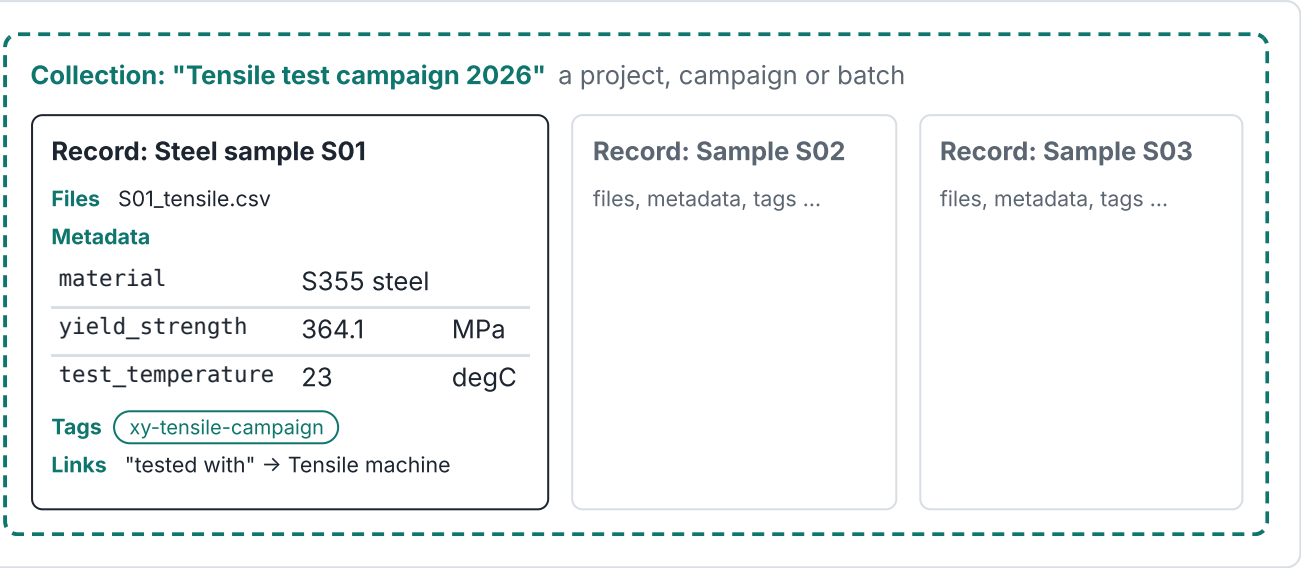
- [ ] **7. Prepare your screen** 2 min
  Test microphone and the meeting chat. If you can, use two screens, or put the meeting and the notebook side by side. Keep Kadi4Mat open in a second browser tab.

> **Something didn't work?** No problem. Send a screenshot of the error to the help address above before the workshop, or join 10 minutes early. Don't skip the checklist: setup problems are much easier to fix before the session than during it.

THE BIG PICTURE

Kadi4Mat in one minute

Kadi4Mat stores research data in a structured, searchable way. Three ideas are enough for today:



TERM	WHAT IT MEANS FOR YOU
Record	One "thing" you want to document: a sample, a measurement, a simulation run. Holds files and metadata.
Metadata	Structured facts about a record: **key** (name), **type** (text, number, ...), **value** and **unit**. Searchable, unlike numbers buried in a file.
Collection	A folder-like group of records, e.g. a project or test campaign.
Tag	A keyword label for finding records later.
Identifier	A unique short name for a record, e.g. `jd-sample-s01`. Unique across all of Kadi4Mat, so we put **your initials** in front.

What is kadi-apy?

A remote control for Kadi4Mat. Anything you can click in the website, kadi-apy can do for you, for one file or for a thousand.

Website
Click by hand. Fine for a few records.

Command line
`kadi-apy records ...`
One command per action. Easy to repeat.

Python
`from kadi_apy import ...`
Loops, calculations, reading your files.

> **Where it saves you time:** uploading a whole folder of instrument files at once; extracting results (e.g. yield strength) into metadata automatically; consistent names and tags for the whole group; finding and downloading data for a report or paper.

TOOLS

Google Colab in one minute

YOU WANT TO ...	DO THIS
Run a cell	Click into it, press `Shift` + `Enter` (or the play button on the left).
Know the order	Always **top to bottom**. Later cells use results of earlier ones.
Do an exercise	Change only lines marked `TODO`, then run the cell.
Get unstuck	Click **Show solution** below the exercise and copy the code.
Read a `!` line	Lines starting with `!` are terminal commands, not Python.
Fix "everything is broken"	**Runtime → Restart session**, then run all cells again from the top (section 0 first).
Understand an error	Read the **last line** of the red message first. Colab may also offer an **Explain error** button.

Python survival kit: the only 6 things you need today

Indentation matters: the spaces at the start of a line tell Python what belongs inside a loop. Keep them exactly as in the template.

1 · Variable: a name for a value

```
PREFIX = "jd"
temperature = 23.0
```

2 · Text with values inside (f-string)

```
f"{PREFIX}-sample-s01"   # -> "jd-sample-s01"
```

3 · List: several things in order

```
samples = ["S01", "S02", "S03"]
```

4 · Dictionary: named fields

```
{"key": "yield_strength", "type": "float",
 "value": 364.1, "unit": "MPa"}
```

5 · For loop: repeat for every item

```
for sample in samples:
    print(sample)  # indented: inside
```

6 · Calling a function / method

```
record.upload_file("S01_tensile.csv")
# object . action ( details )
```

During the online session
Post a quick message in the chat when you finish an exercise.
Stuck for more than 2 minutes? Write in the chat; a helper will take you to a breakout room.
No question is too basic.

After the workshop
Docs: kadi-apy.readthedocs.io
Material: github.com/msishuvo/thematic-workshop-kadiAPY
Using real lab data? Check your institute's data policy before uploading it to Colab, or install kadi-apy locally: `pip install kadi-apy`.

KEEP THIS OPEN DURING THE WORKSHOP

kadi-apy cheat sheet

Command line (in Colab, put `!` in front)

```
# create a record
kadi-apy records create -i jd-sample-s01 -t "Sample S01"
# upload a file (or a whole folder)
kadi-apy records add-files -R jd-sample-s01 -n data/tensile_tests/S01_tensile.csv
# add metadata with type and unit
kadi-apy records add-metadatum -R jd-sample-s01 -m specimen_length -v 80 -t float -u mm
# add a tag
kadi-apy records add-tag -R jd-sample-s01 -t jd-tensile-campaign
# show metadata (-m) and file list (-l)
kadi-apy records show-info -R jd-sample-s01 -m -l
# find records by tag / list all commands
kadi-apy records get-records -t jd-tensile-campaign
kadi-apy records --help
```

Python

```
from kadi_apy import KadiManager
manager = KadiManager()                        # your login (reads the config)

# get the record, or create it if it does not exist yet
record = manager.record(identifier="jd-sample-s01", title="Sample S01", create=True)

# metadata: a list of dictionaries (key, type, value, unit)
record.add_metadata([{"key": "specimen_length", "type": "float",
                      "value": 80.0, "unit": "mm"}], force=True)

record.upload_file("data/tensile_tests/S01_tensile.csv", force=True)   # upload a file
record.add_tag("jd-tensile-campaign")                    # add a tag

# put the record into a collection
collection = manager.collection(identifier="jd-campaign", title="Campaign", create=True)
record.add_collection_link(collection.id)
```

If you see this error ...

MESSAGE (LAST LINE)	USUALLY MEANS / FIX
NameError: name 'manager' is not defined	An earlier cell was not run (often after a restart). Run section 0 again, top to bottom.
Could not read the secret KADI_PAT	Secret name misspelled, or **Notebook access** is off. Check the key icon in the sidebar.
401 / "invalid access token"	Token copied incompletely or expired. Create a new one and update the secret.
identifier ... already in use	Someone else owns that identifier. Check that your prefix is set to **your** initials.
IndentationError	Spaces at the start of a line don't match. Copy the line from the solution.
SyntaxError	Usually a missing quote `"`, bracket `)` or comma. Compare with the solution.